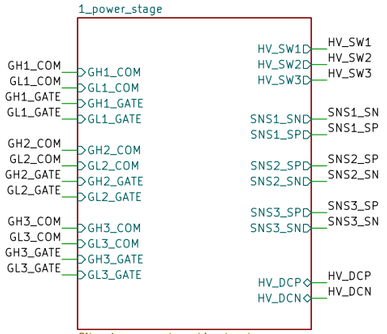
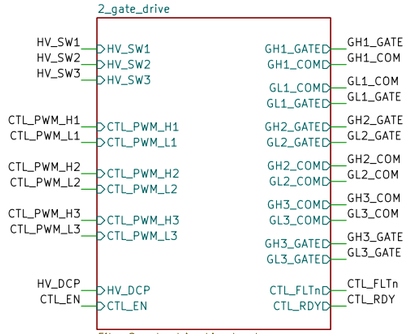
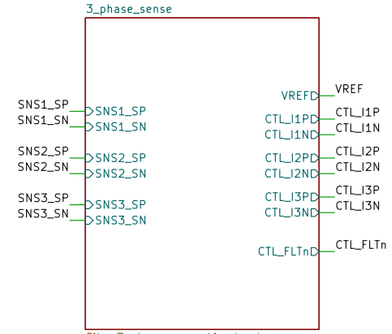
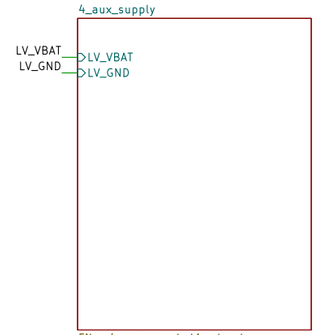
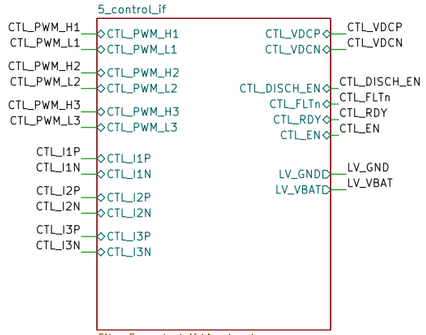
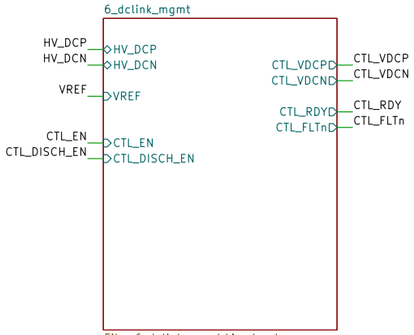



SYSTEM REQUIREMENTS (report, Table 'Requirements on the controlling system')
R1–R12 and R14 are mandatory: the board's verification depends on them.
R13 is recommended: it adds margin the verification does not count on.

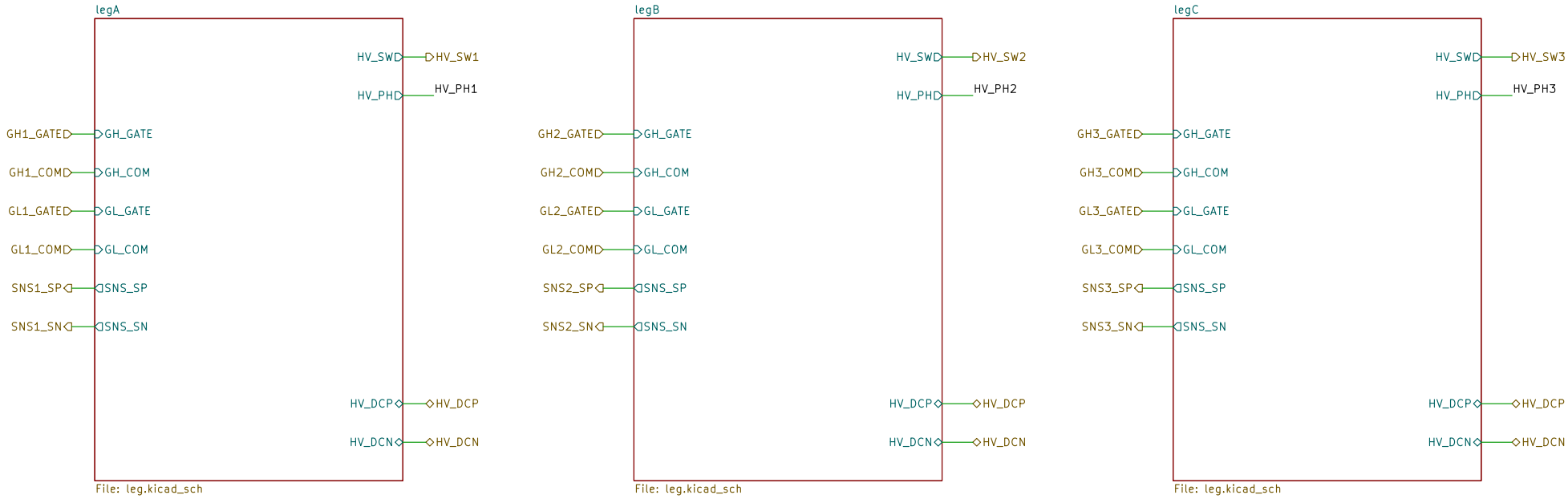
R1 Six PWM signals, complementary in pairs, 3.3 V CMOS, dead time >= 260 ns generated by the peripheral
R2 Enable low in every indeterminate state: reset, boot, watchdog
R3 A DESAT fault is reset only by holding enable low for > 1 us
R4 Modulation enabled only after the ready line goes high (all UVLO released)
R5 Main contactor closes on the measured DC–link voltage, not on a timer
R6 Differential ADC inputs, sampled synchronously with the carrier peak/valley
R7 ADC trigger advanced by 3.1 us
R8 Derating from the heat–sink thermistor
R9 Interlock loop supervision and contactor command by the vehicle
R10 Galvanic isolation in the charging path (on–board AC or in the DC station)
R11 Source impedance at 20 kHz: L >= 1.5 uH or R >= 50 mOhm
R12 Max continuous battery voltage 900 V; up to 1000 V only with the bridge disabled and for < 30 min a day
R13 (recommended) Below a few hertz of output frequency, DPWM–MAX: clamp to the positive rail the phase carrying the largest positive current
R14 Coolant at the rail inlet: 50/50 water–glycol, <= 65 C, >= 10 l/min

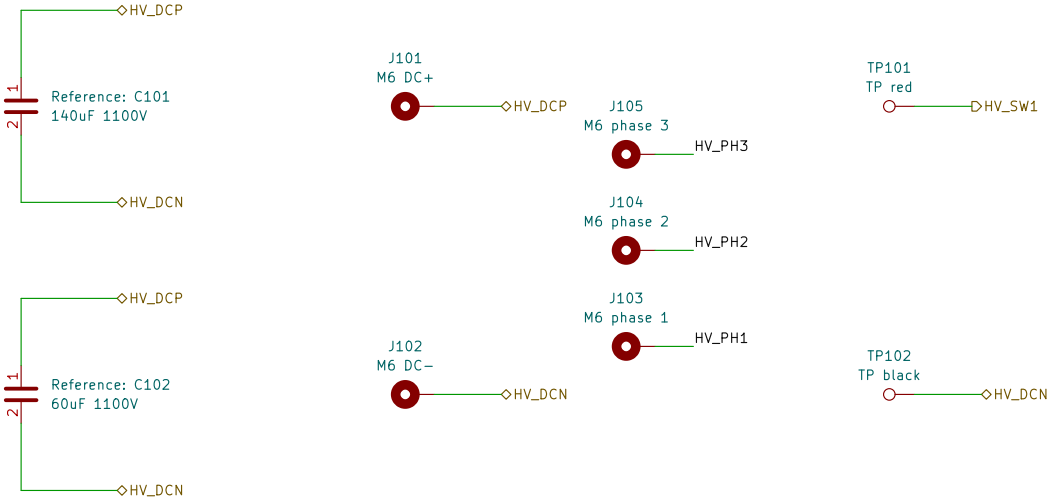
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Sheet: /
File: PECD_P7.kicad_sch

***Title: Root sheet: blocks, interface and system requirements***

| | | |
|---|---|---|
| Size: A3 | Date: 2026–10–11 | **Rev: 1.2** |
| KiCad E.D.A. 10.0.6 | | Id: 1/20 |



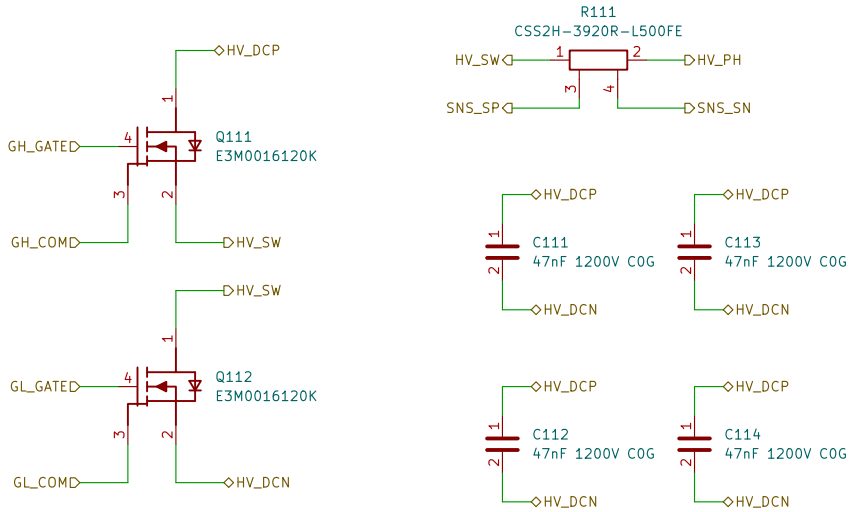




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Sheet: /1_power_stage/
File: 1_power_stage.kicad_sch

***Title: Power stage: DC−link and three legs***

| Size: A4 | Date: 2026−10−11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/20 |

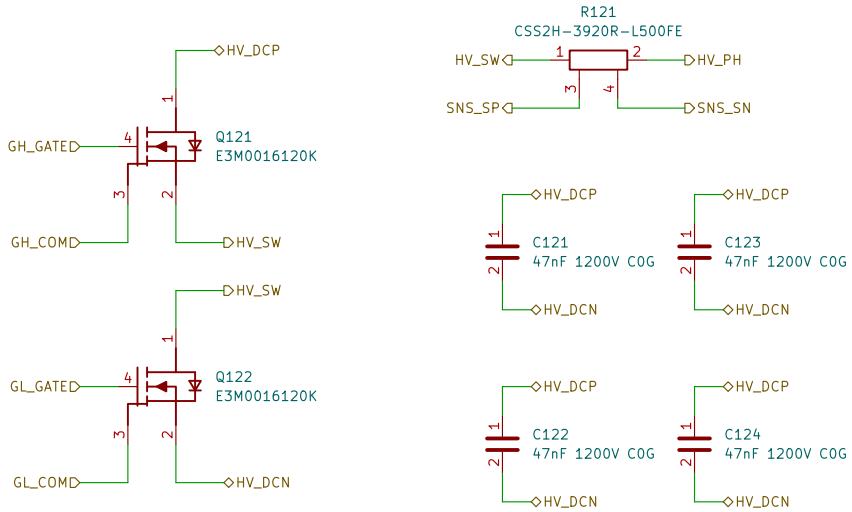



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Sheet: /1_power_stage/legA/
File: leg.kicad_sch

***Title: Half-bridge leg (3 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 8/20 |

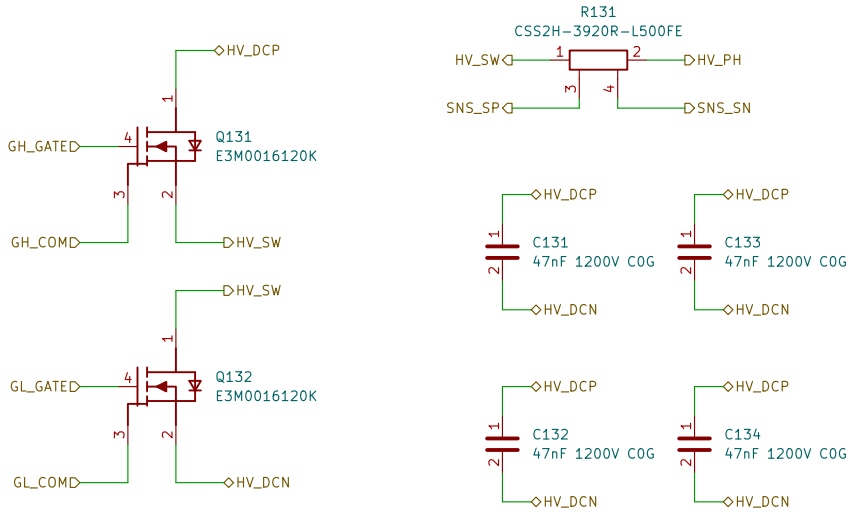



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Sheet: /1_power_stage/legB/
File: leg.kicad_sch

***Title: Half-bridge leg (3 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 9/20 |




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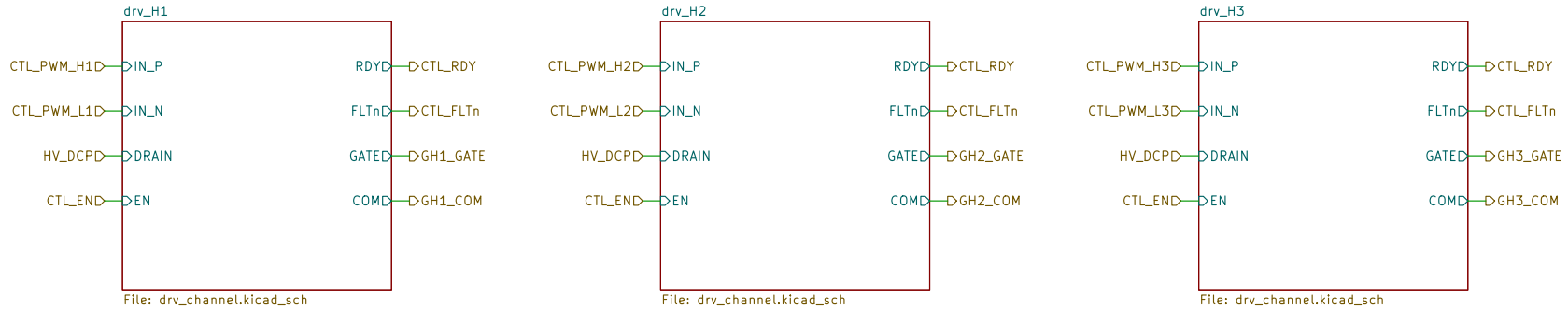
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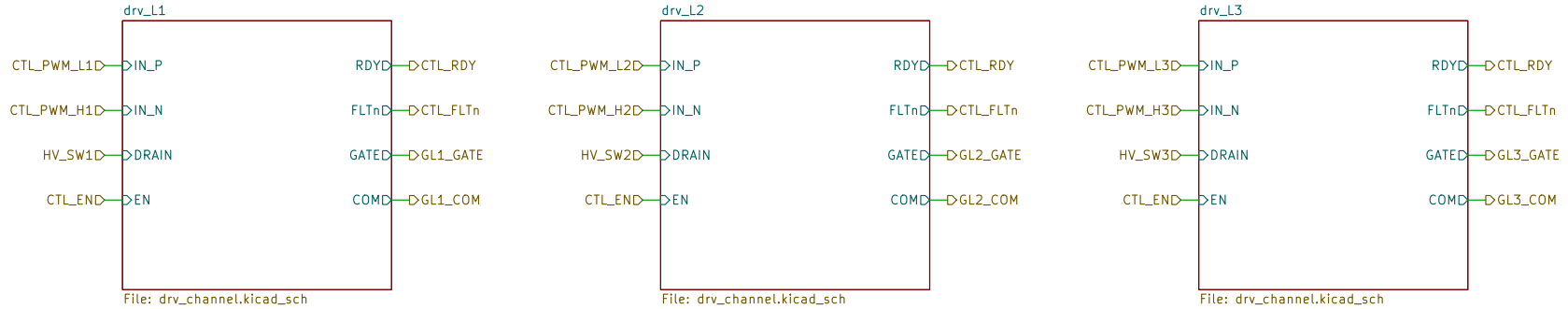
Sheet: /1_power_stage/legC/

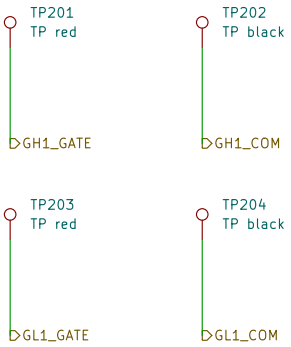
File: leg.kicad_sch

***Title: Half-bridge leg (3 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 10/20 |



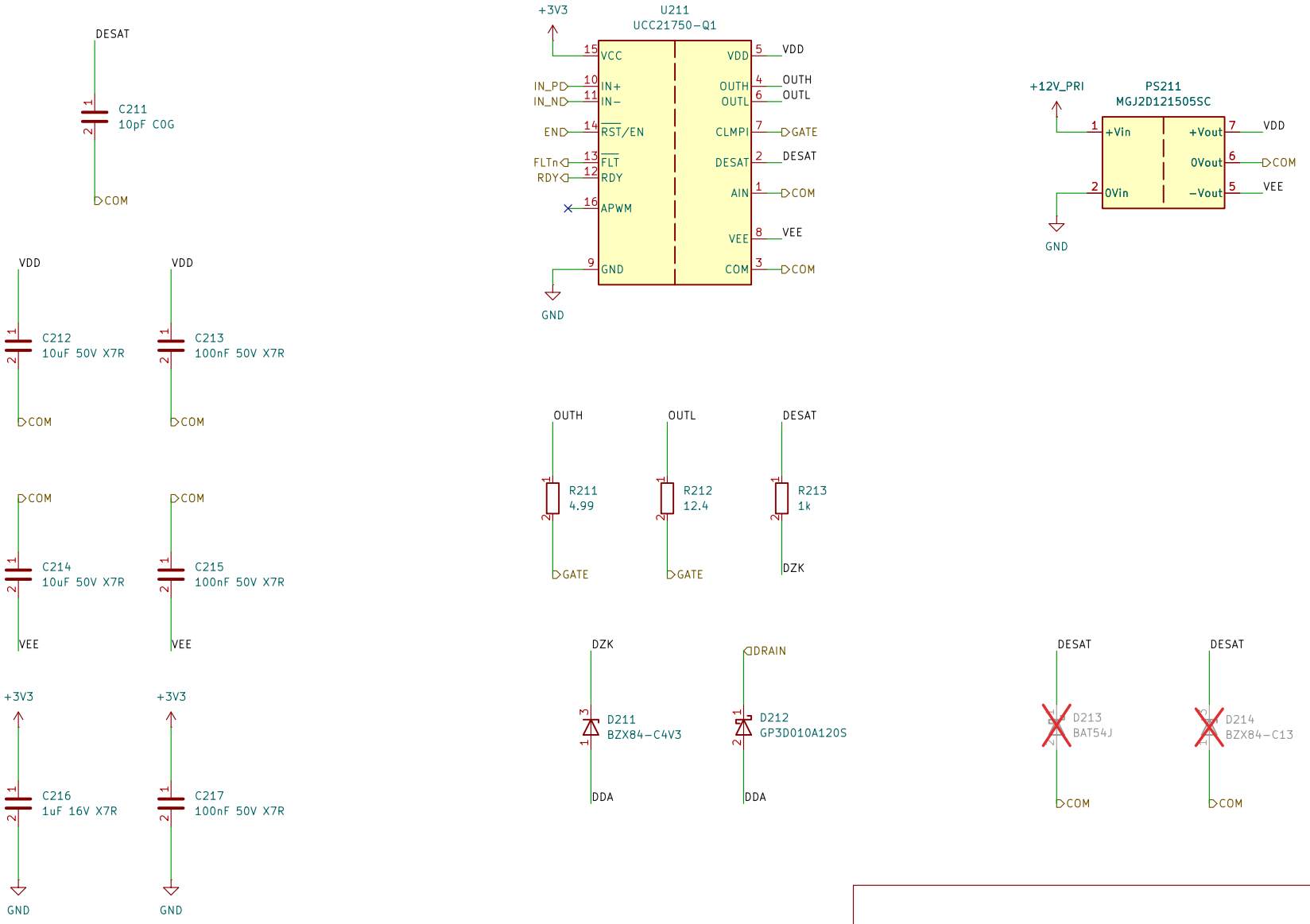




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Sheet: /2_gate_drive/
File: 2_gate_drive.kicad_sch

***Title: Gate drive: seven channels and interlock***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

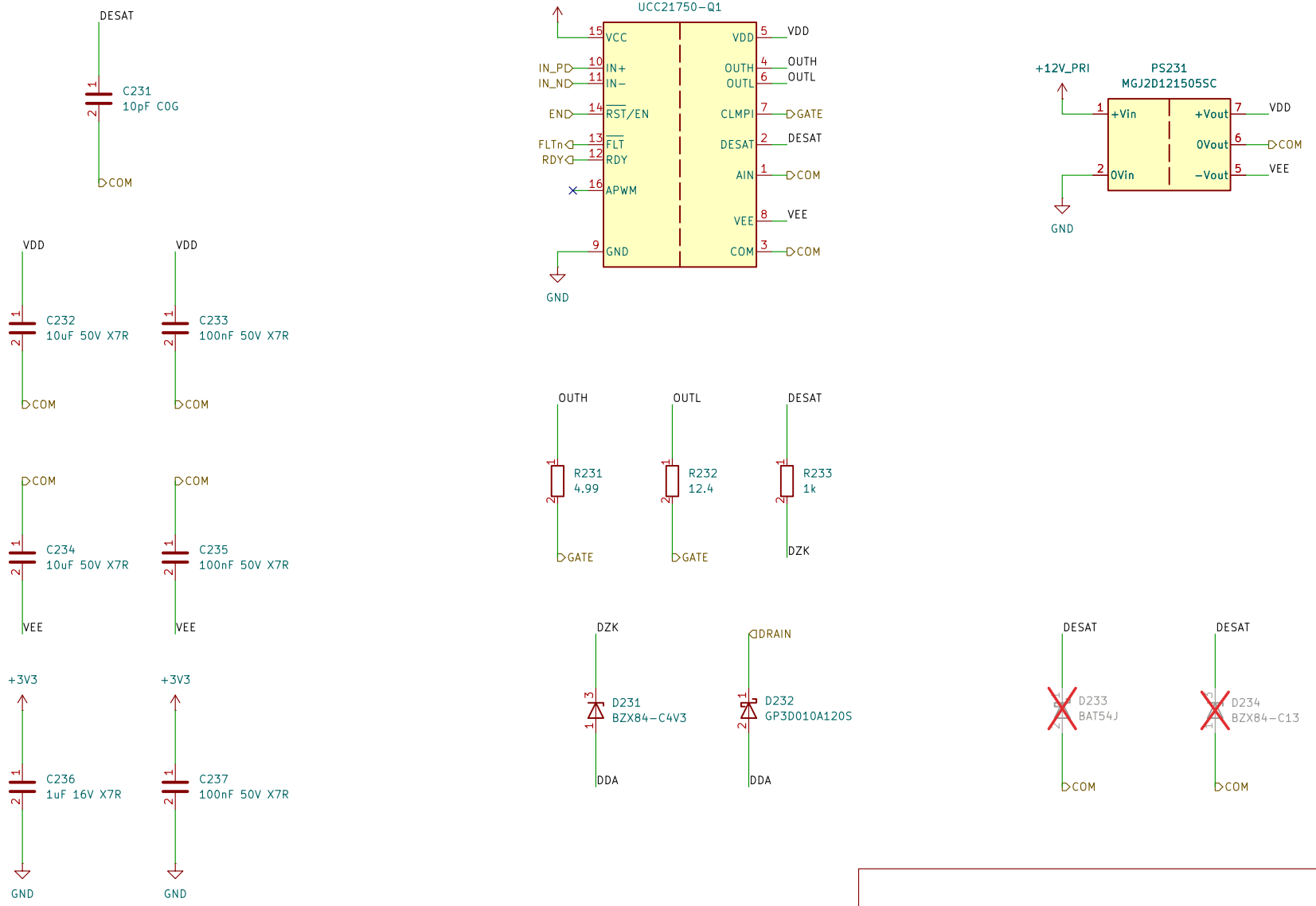
**University of Bologna — Automotive Power Electronics Circuit Design, Project 7**

Sheet: /2_gate_drive/drv_H1/

File: drv_channel.kicad_sch

***Title: Gate-drive channel (7 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 11/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

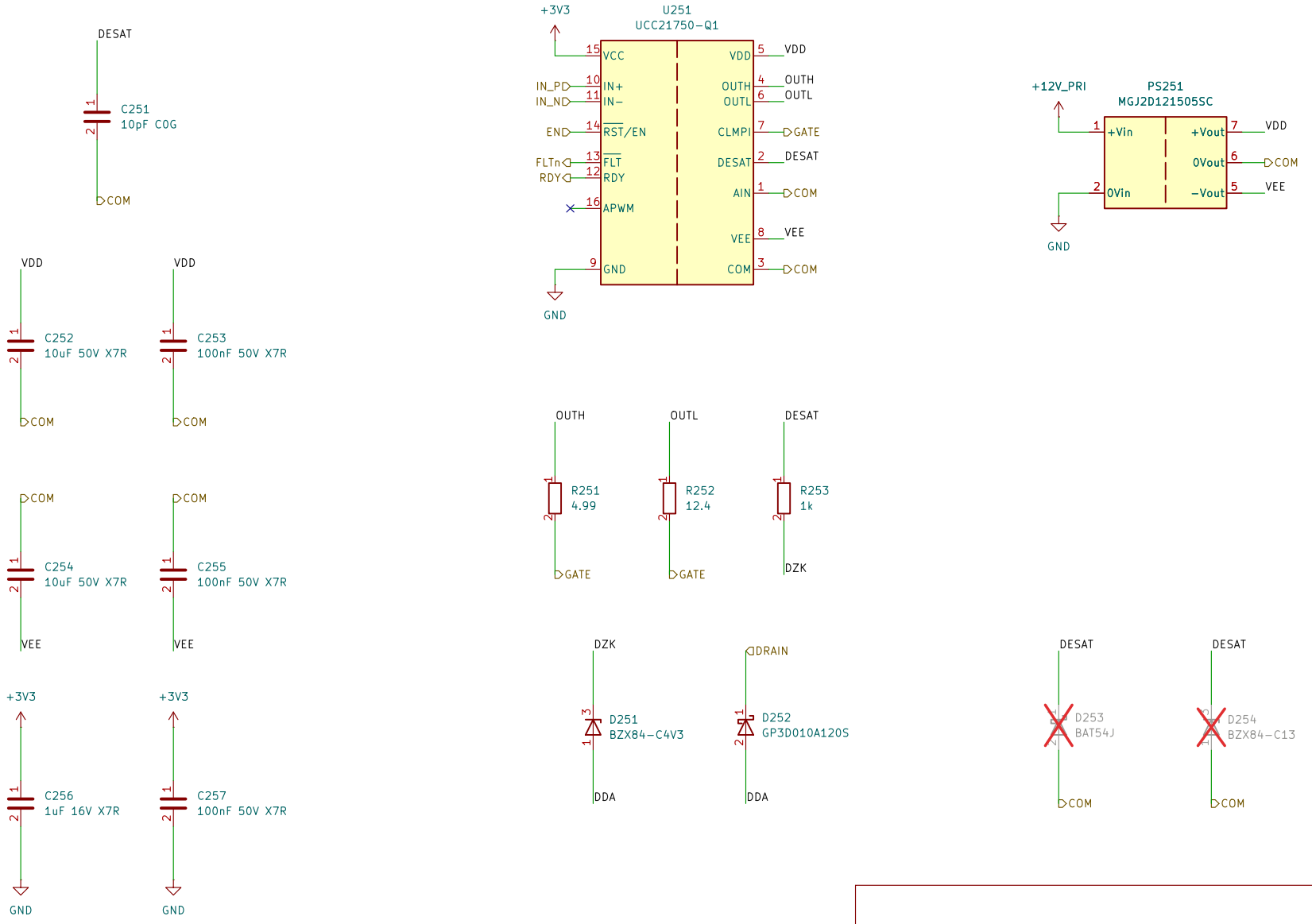
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Sheet: /2_gate_drive/drv_H2/

File: drv_channel.kicad_sch

***Title: Gate-drive channel (7 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 12/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

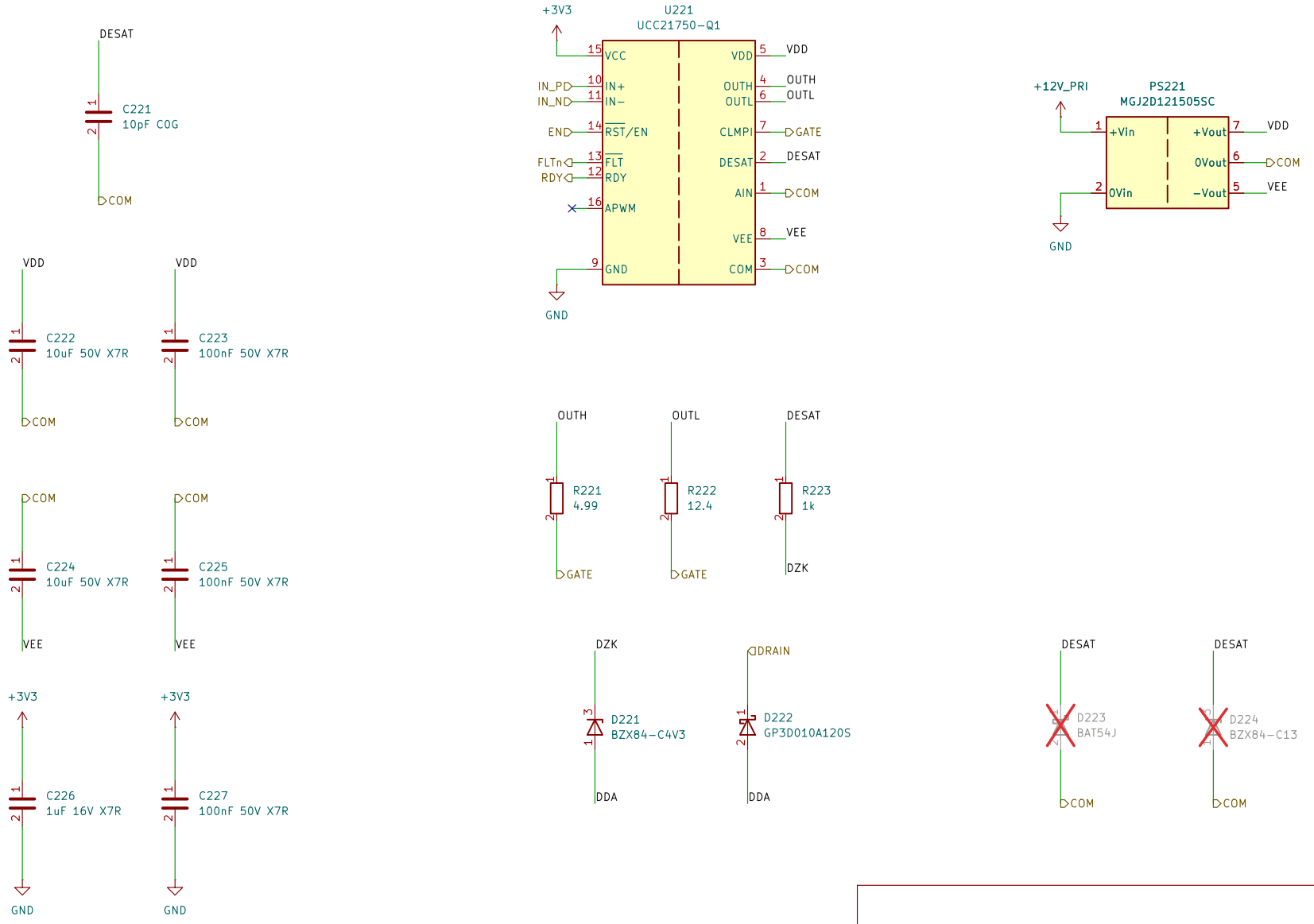
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Sheet: /2_gate_drive/drv_H3/

File: drv_channel.kicad_sch

***Title: Gate-drive channel (7 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 13/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

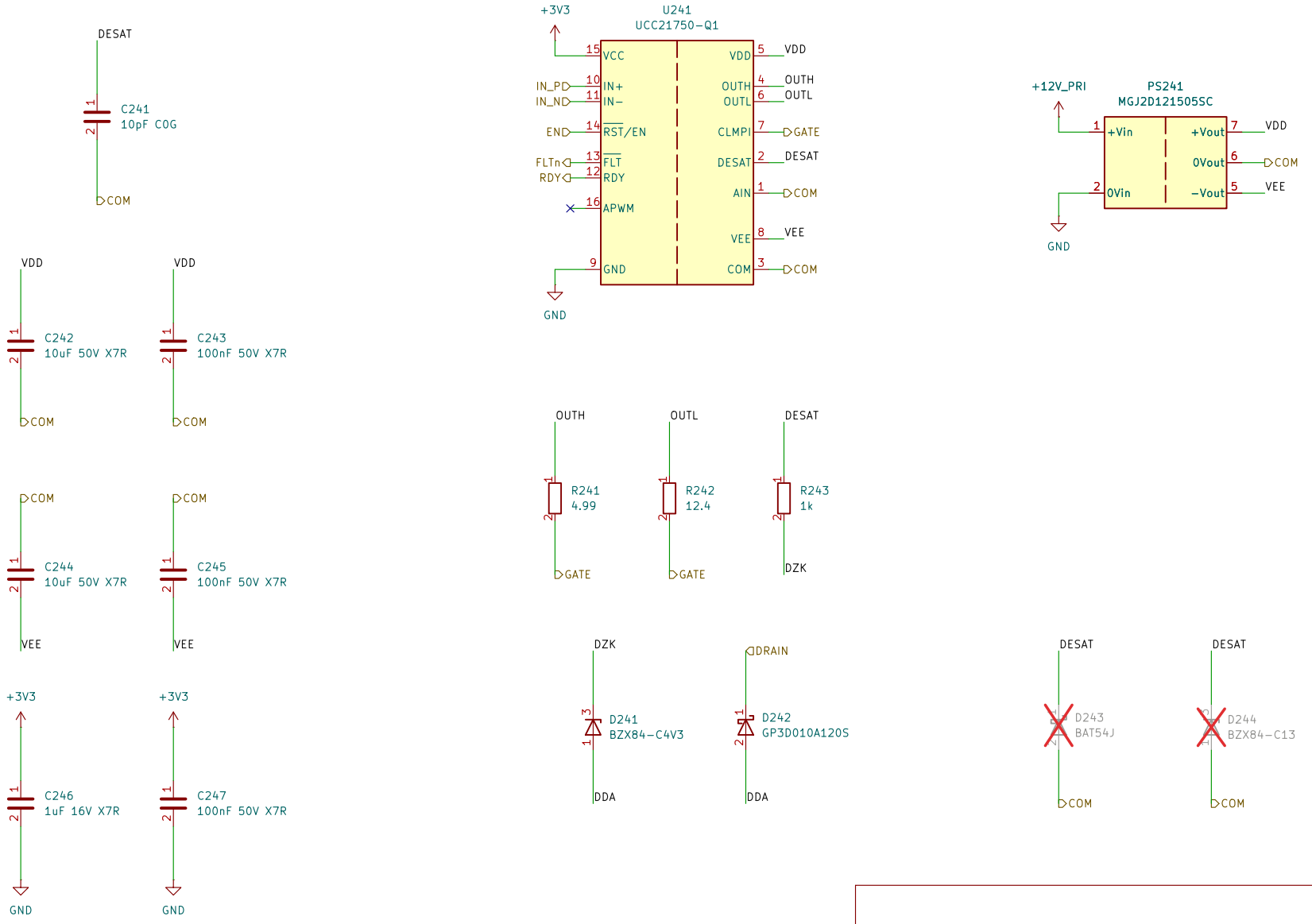
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Sheet: /2_gate_drive/drv_L1/

File: drv_channel.kicad_sch

***Title: Gate-drive channel (7 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 14/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

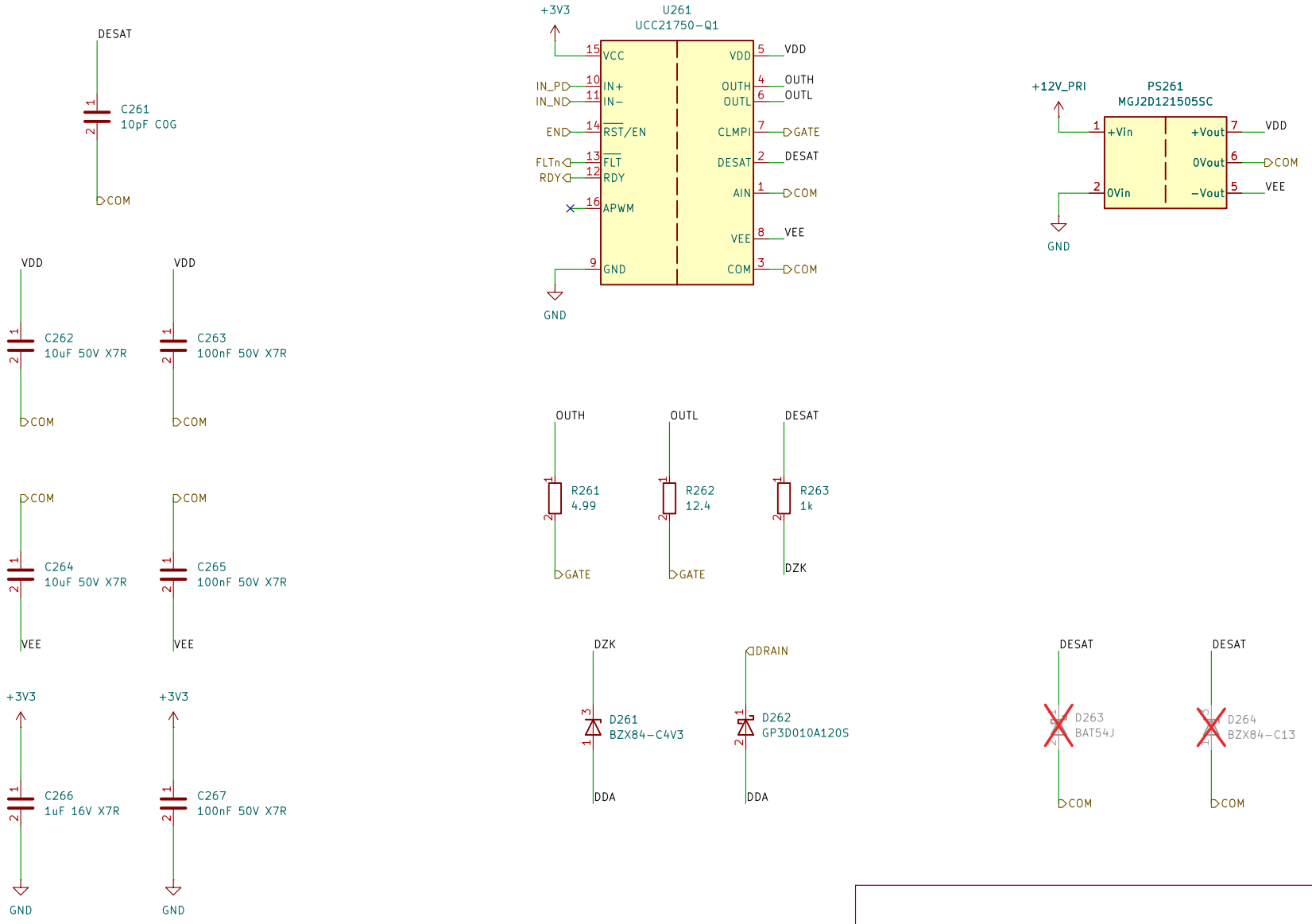
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Sheet: /2_gate_drive/drv_L2/

File: drv_channel.kicad_sch

***Title: Gate-drive channel (7 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 15/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

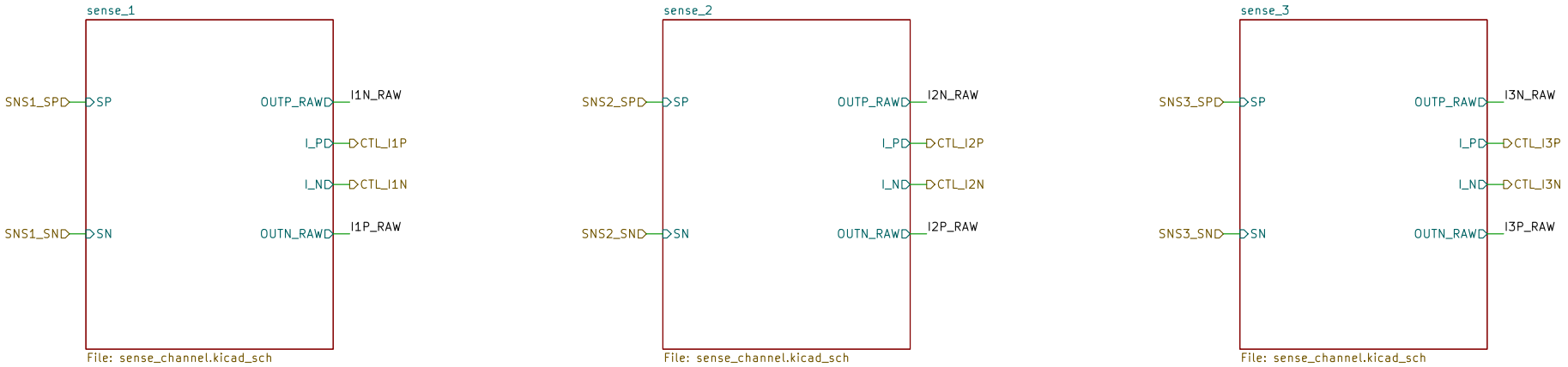
**University of Bologna – Automotive Power Electronics Circuit Design, Project 7**

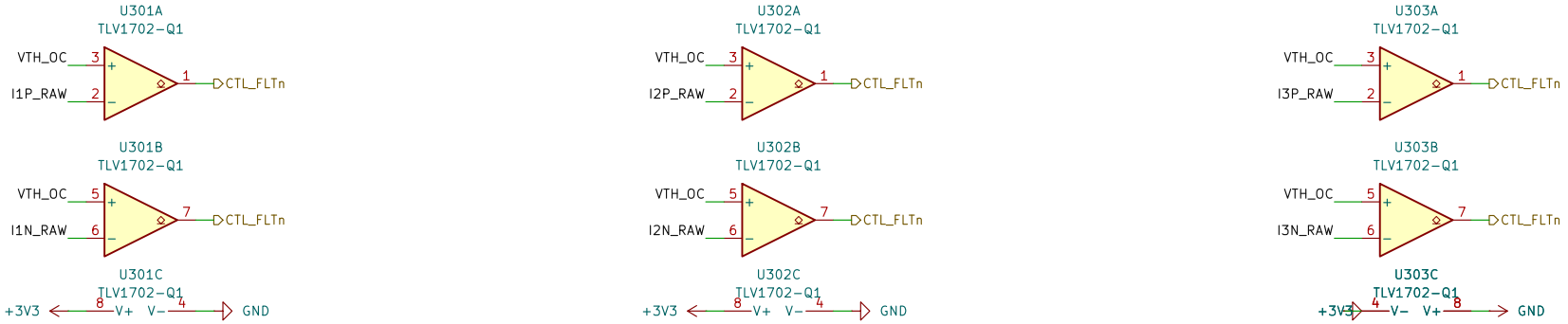
Sheet: /2_gate_drive/drv_L3/

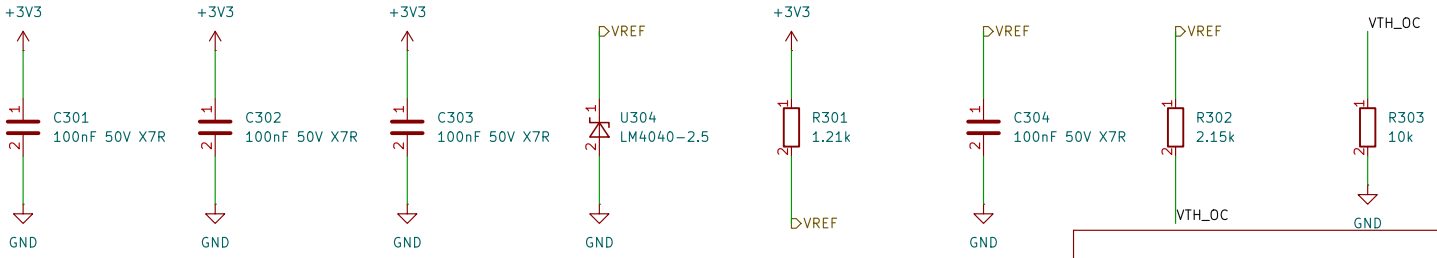
File: drv_channel.kicad_sch

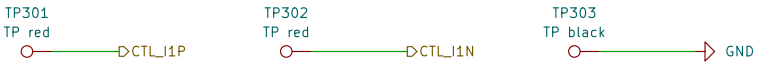
***Title: Gate-drive channel (7 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 16/20 |



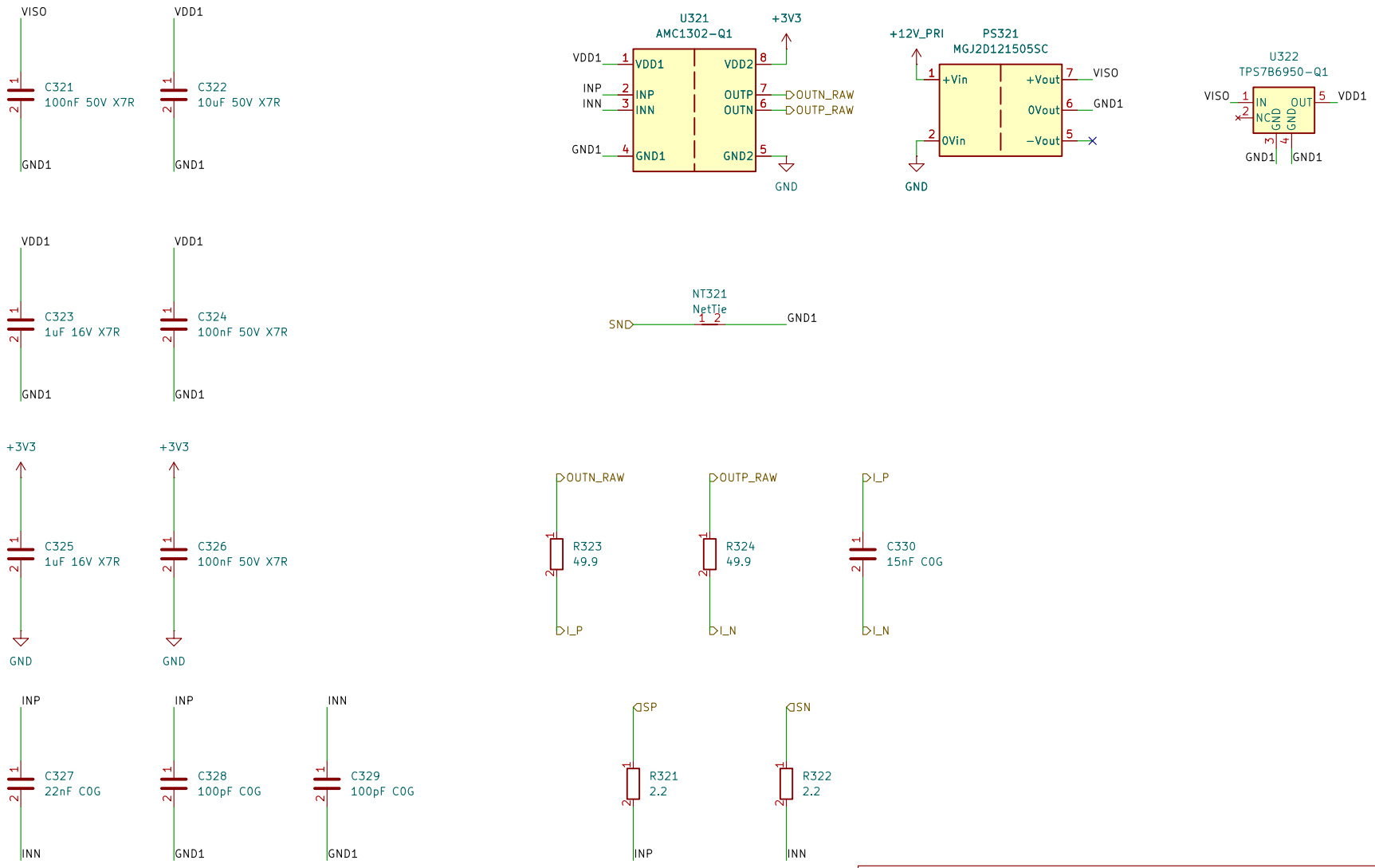




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Sheet: /3_phase_sense/
File: 3_phase_sense.kicad_sch

***Title: Phase current sensing and overload comparators***

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
| KiCad E.D.A. 10.0.6 | | Id: 4/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

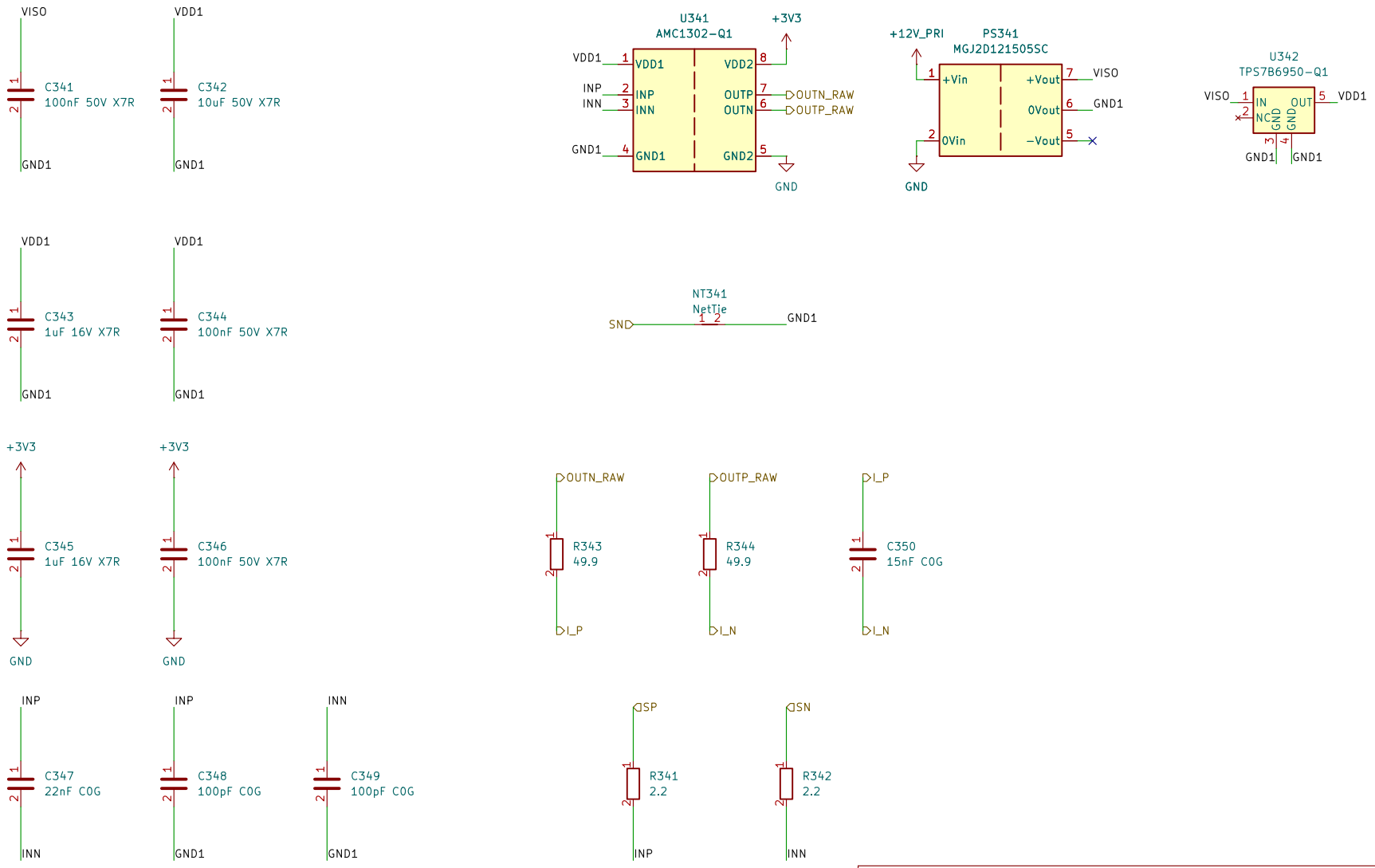
**University of Bologna — Automotive Power Electronics Circuit Design, Project 7**

Sheet: /3_phase_sense/sense_1/

File: sense_channel.kicad_sch

***Title: Phase current sensing channel (3 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 17/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

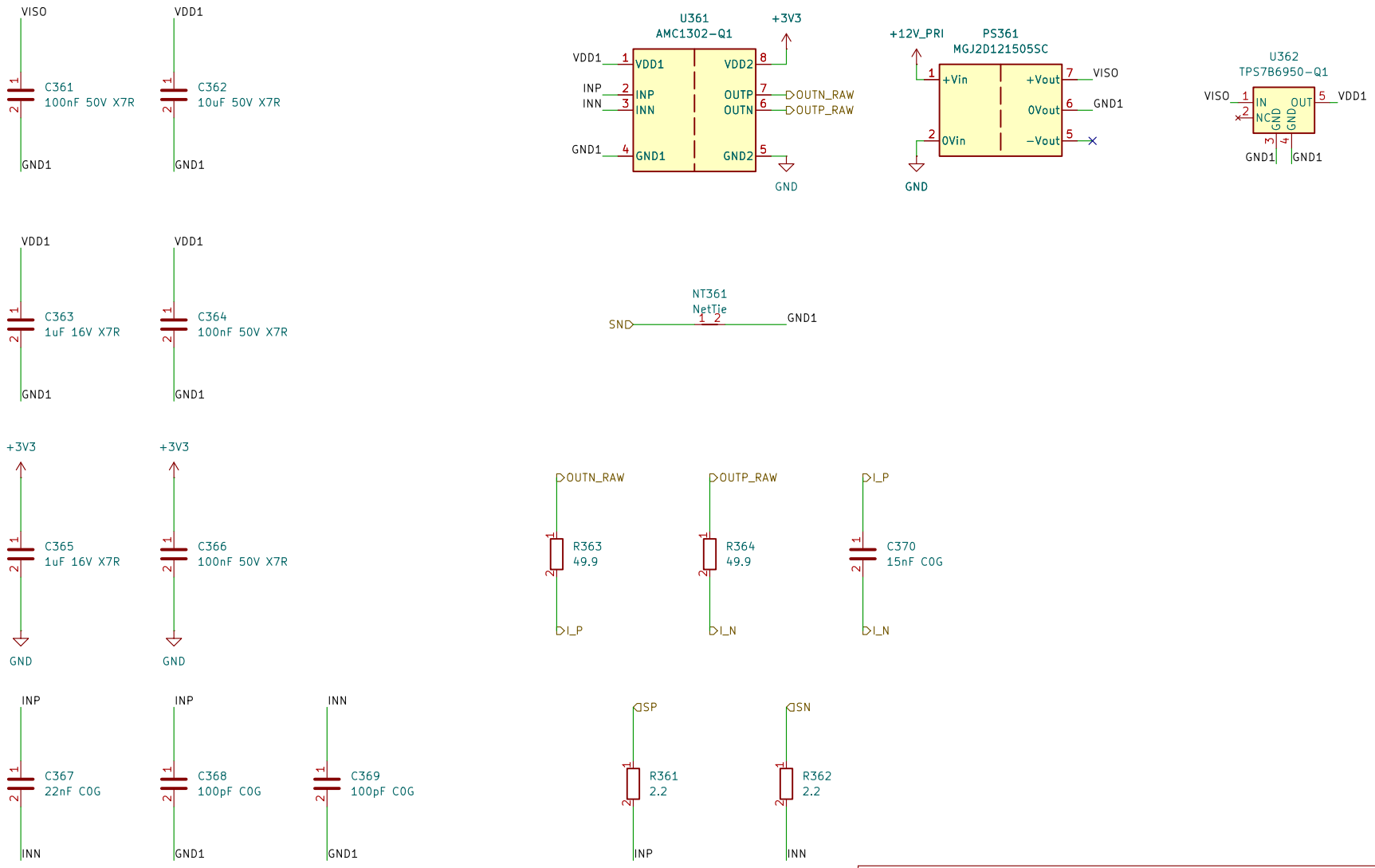
**University of Bologna – Automotive Power Electronics Circuit Design, Project 7**

Sheet: /3_phase_sense/sense_2/

File: sense_channel.kicad_sch

***Title: Phase current sensing channel (3 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 18/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

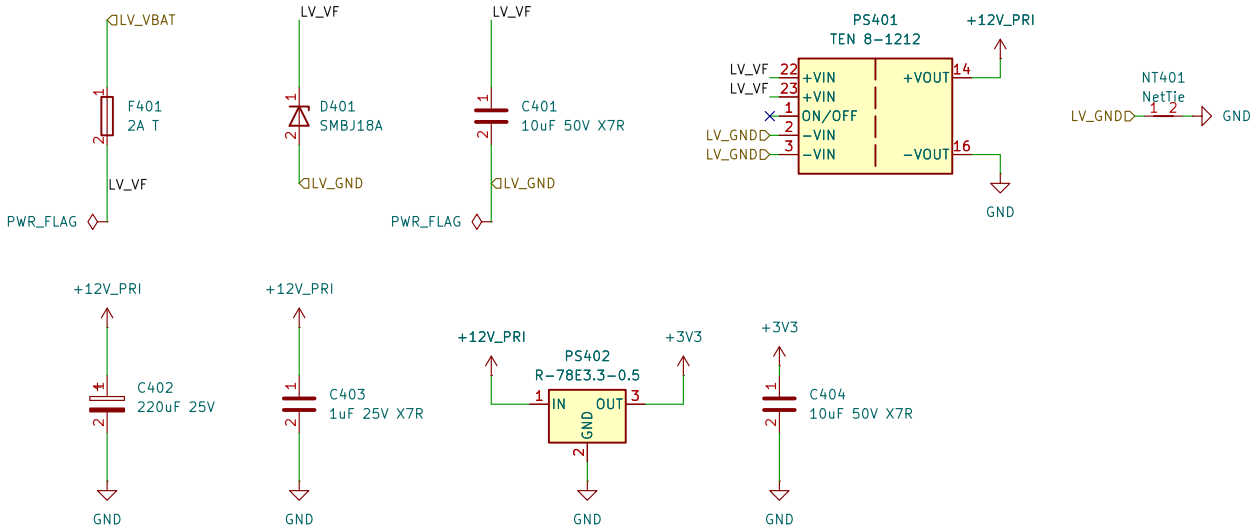
**University of Bologna — Automotive Power Electronics Circuit Design, Project 7**

Sheet: /3_phase_sense/sense_3/

File: sense_channel.kicad_sch

***Title: Phase current sensing channel (3 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 19/20 |

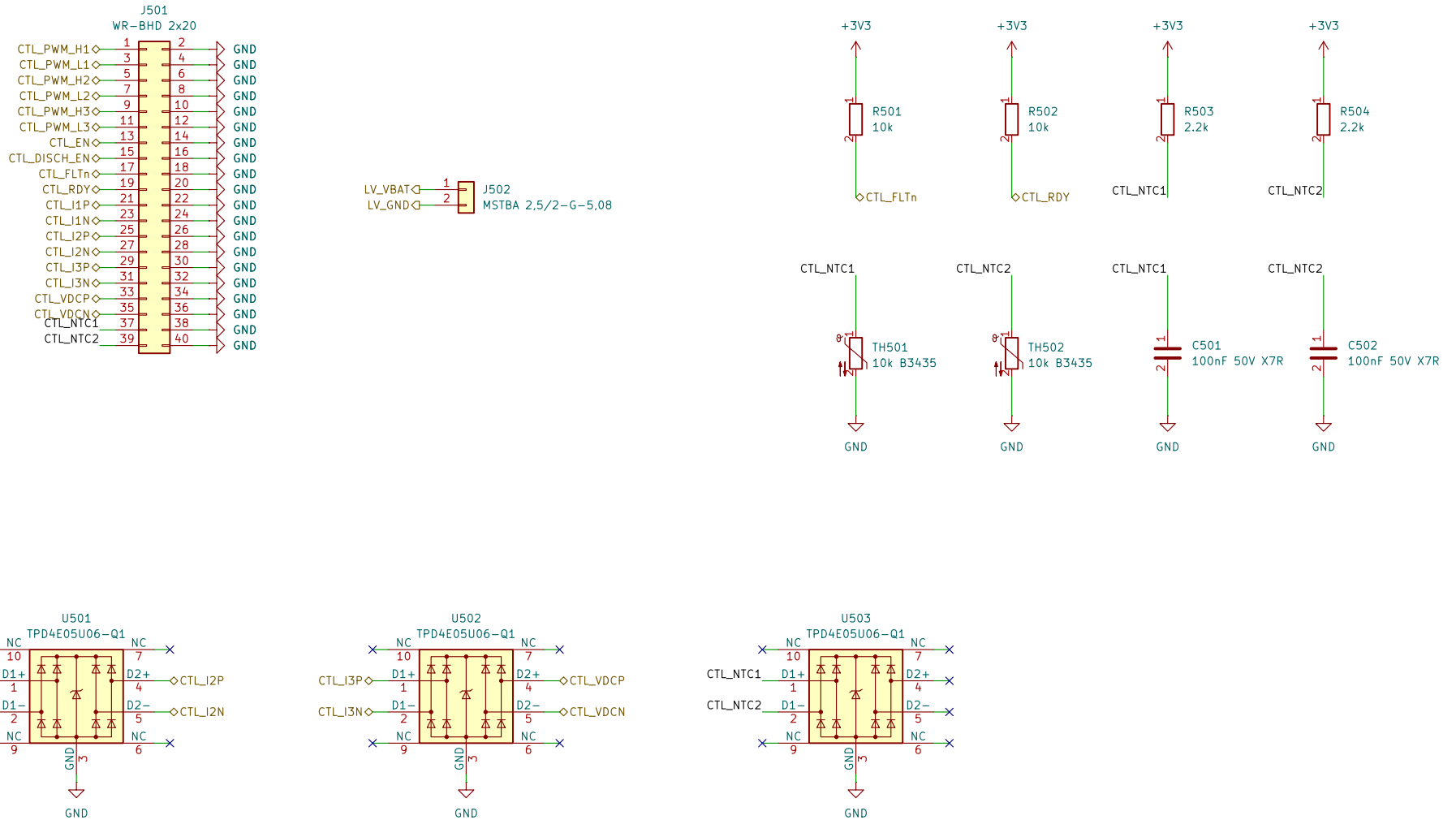



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Sheet: /4_aux_supply/
File: 4_aux_supply.kicad_sch

***Title: Auxiliary supply: 12 V regulated and 3.3 V***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

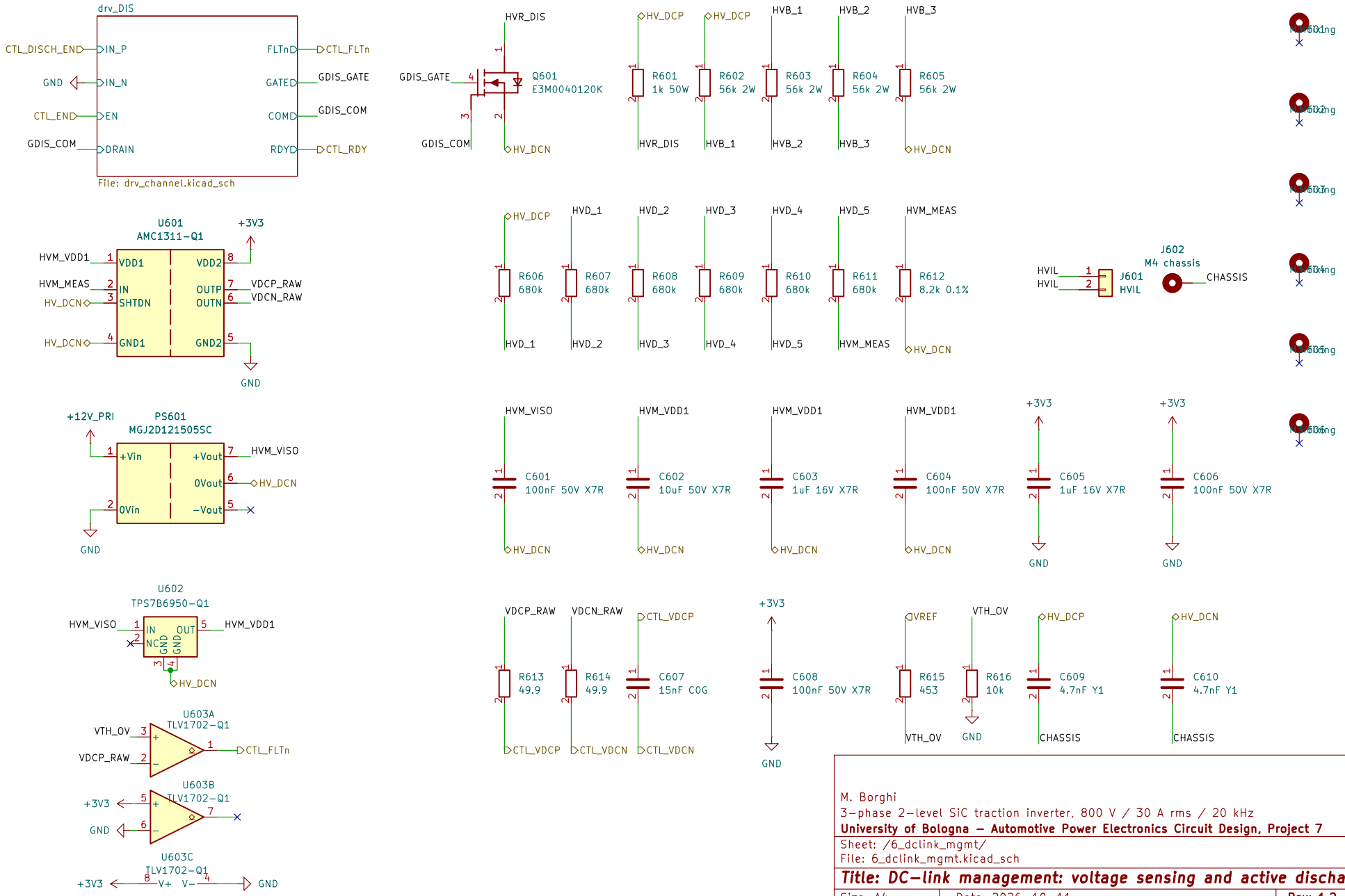
**University of Bologna — Automotive Power Electronics Circuit Design, Project 7**

Sheet: /5_control_if/

File: 5_control_if.kicad_sch

***Title: Control interface: connector, filters, thermistors***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 6/20 |




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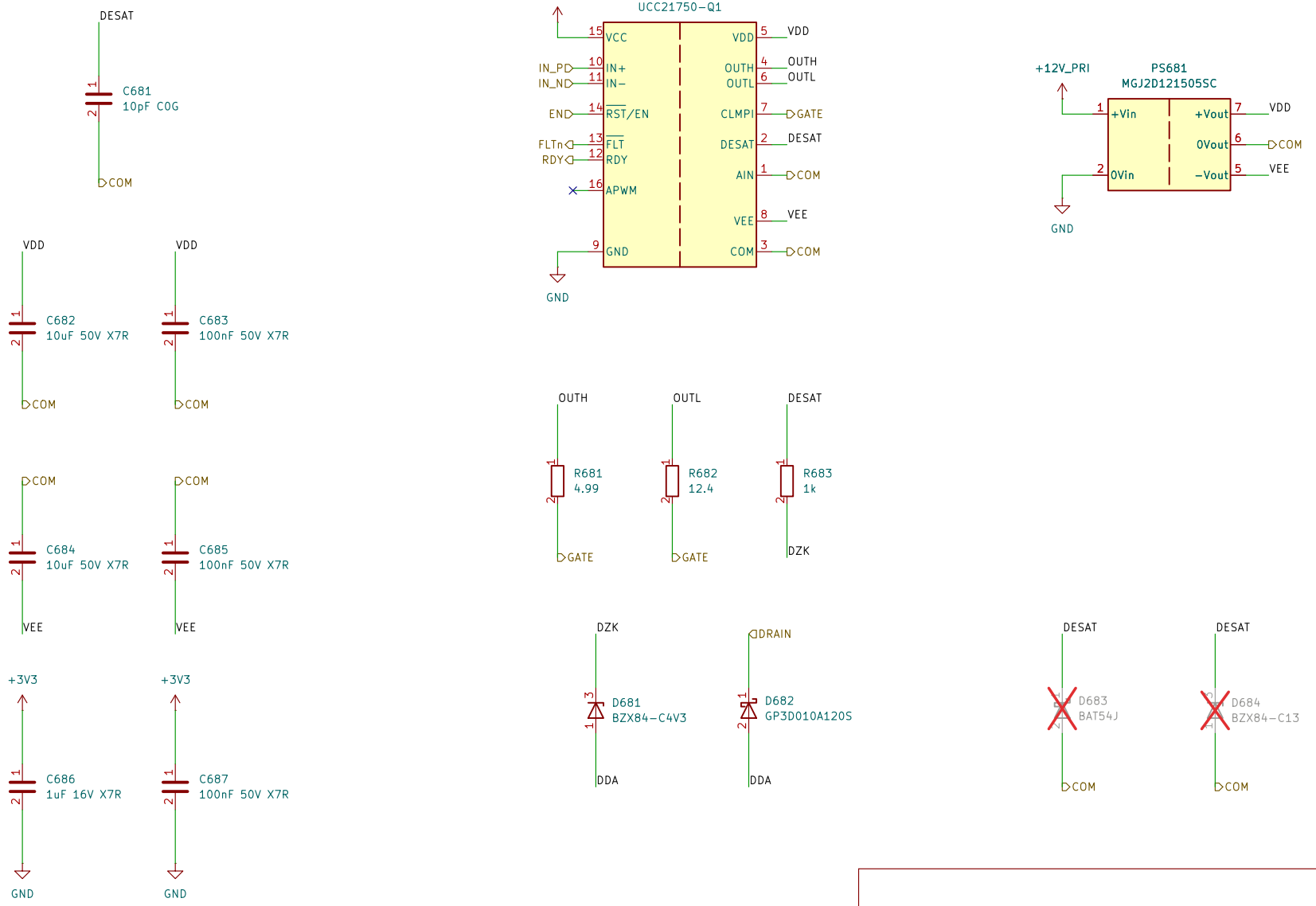
**University of Bologna – Automotive Power Electronics Circuit Design, Project 7**

Sheet: /6_dclink_mgmt/

File: 6_dclink_mgmt.kicad_sch

***Title: DC-link management: voltage sensing and active discharge***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 7/20 |




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3-phase 2-level SiC traction inverter, 800 V / 30 A rms / 20 kHz

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Sheet: /6_dclink_mgmt/drv_DIS/

File: drv_channel.kicad_sch

***Title: Gate-drive channel (7 instances)***

| Size: A4 | Date: 2026-10-11 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 20/20 |